

Enterprise Learning Platform

with Skill and Career Guidance System

Milestone 2 — Project Documentation

From Real Dashboard to a Working Learning Platform

Team C

Prepared for team review

1. Overview — Where We Resumed From

Milestone 1 delivered the foundation of the platform: a working Google OAuth2 + JWT login flow, a MySQL database with a single users table, and a fully designed themed landing page. No dashboard, no courses, and no learning content existed yet — the project was, by design, an authentication skeleton with a UI shell around it.

Milestone 2 picks up from that exact point and turns the platform into a functioning learning system: a real dashboard backed by a real database, a browsable course catalog with a working enrollment flow, actual lesson content (videos, readings, and server-graded quizzes), forgot/reset password via real email, expanded signup details, and a full account settings page.

Every feature below was built, tested against the running backend, and pushed to the team's shared repository.

1.1 What Was Already Done (Milestone 1 Recap)

- Google OAuth2 login flow (Spring Security + Google Cloud Console), fully working end to end
- JWT generation, validation, and route protection (JwtUtil, JwtAuthenticationFilter)
- MySQL database with a single users table (id, name, email, googleId, password, profilePicture)
- Themed React landing page (Hero, Routes, Instruments, TrailLog, Navbar, Connect)
- Global auth state via AuthContext (token, login, logout, isAuthenticated)

2. Milestone 2 — Feature Summary

2.1 Real Dashboard (Replacing Mock Data)

The dashboard was originally built against hardcoded mock data. It now pulls 100% real data from the backend: enrollment stats, a 7-day activity chart, per-category skill progress, and achievements computed live from actual user behavior rather than hardcoded.

- Streak, weekly activity, and “hours this week” are calculated from a real daily activity log
- Achievements (First Summit, 7-Day Streak, Basecamp Reached, Full Ascent) are evaluated dynamically against real enrollment/completion counts
- Skill progress is grouped and averaged live by course category across all of a user's enrollments

2.2 Course Catalog & Real Enrollment

A full course browsing page was built, replacing the earlier state where courses could only be seeded via a developer-only endpoint. Users can now see every course, whether they're enrolled, and click a real Enroll button that creates a database row.

2.3 Real Lesson Content — Video, Reading, and Quiz

This is the largest single addition in Milestone 2. Each course now has a real ordered list of lessons of three types:

- Video lessons — an embedded, playable video with a “Mark as complete” button.

- Reading lessons — text content with a “Mark as complete” button.
- Quiz lessons — multiple-choice questions, graded entirely server-side (correct answers never sent to the browser), passing at 70% or higher.

A dedicated course player page was built: a lesson sidebar with completion checkmarks, and a main panel that renders whichever lesson is selected. Completing a lesson or passing a quiz recalculates real enrollment progress from actual completed lessons — replacing an earlier, simpler button that blindly added one unit per click.

2.4 Forgot / Reset Password (Real Email)

- User requests a reset from a new “Forgot password?” link on the login page
- Backend generates a random token with a 30-minute expiry and emails a real reset link to the user's inbox
- A dedicated Reset Password page reads the token from the URL and lets the user set a new password
- Google-only accounts (no password set) are told directly to use Google sign-in instead, matching the message already used at login

2.5 Personal Details at Signup

The signup form was extended beyond name/email/password to collect phone number, date of birth, gender, and location — all optional, all saved to the user's record.

2.6 Account / Profile Settings Page

A new profile page was built: view and edit personal details, see enrollment stats and member-since date, and change password (hidden entirely, with an explanation, for Google-only accounts).

3. Database Changes

All new tables were created automatically by Hibernate — no manual SQL was written. The existing users table was extended with new columns rather than replaced, so no data was lost at any point.

3.1 New Tables Created

Table	Purpose	Key Columns
courses	Course catalog	id, title, category, total_units
enrollments	Links a user to a course + tracks progress	id, user_id, course_id, units_completed, enrolled_at, last_accessed
daily_activity	One row per user per day; powers streak + weekly chart	id, user_id, activity_date, units_completed, hours_spent
lessons	Individual video/reading/quiz units within a course	id, course_id, title, order_index, type, video_url, content

Table	Purpose	Key Columns
quiz_questions	Multiple-choice questions belonging to a quiz lesson	id, lesson_id, question_text, option_a/b/c/d, correct_option
lesson_progress	Tracks per-user, per-lesson completion	id, user_id, lesson_id, completed, completed_at

3.2 Columns Added to the Existing users Table

Column	Purpose
phone_number	Collected at signup (optional)
date_of_birth	Collected at signup (optional)
gender	Collected at signup (optional)
location	Collected at signup (optional)
reset_token	Random token issued when a password reset is requested
reset_token_expiry	30-minute expiry timestamp for the reset token
created_at	Auto-set on account creation; powers “Member since” on the profile page

4. New & Modified Backend Files

4.1 Dashboard & Course Data

- model: Course.java, Enrollment.java, DailyActivity.java
- repository: CourseRepository.java, EnrollmentRepository.java, DailyActivityRepository.java
- service: DashboardService.java
- controller: DashboardController.java, CourseController.java

4.2 Lessons & Quizzes

- model: Lesson.java, LessonType.java, QuizQuestion.java, LessonProgress.java
- repository: LessonRepository.java, QuizQuestionRepository.java, LessonProgressRepository.java
- service: LessonService.java
- controller: LessonController.java
- dto: QuizSubmissionRequest.java

4.3 Forgot / Reset Password

- service: EmailService.java (Spring Mail via SMTP)
- controller: AuthController.java — added forgot-password and reset-password endpoints

- model: User.java — added resetToken, resetTokenExpiry
- repository: UserRepository.java — added findByResetToken
- pom.xml — added spring-boot-starter-mail
- application.properties — added spring.mail.* configuration and app.frontend-url

4.4 Signup Personal Details & Profile Settings

- model: User.java — added phoneNumber, dateOfBirth, gender, location, createdAt
- dto: AuthRequest.java, UpdateProfileRequest.java
- controller: AuthController.java (register), AccountController.java (profile view, edit, password change)

5. API Endpoints Added

5.1 Authentication (Public)

Method	Endpoint	Description
POST	/api/auth/register	Creates an account with personal details
POST	/api/auth/login	Email/password login
POST	/api/auth/forgot-password	Emails a reset link if the account has a password
POST	/api/auth/reset-password	Sets a new password using a valid token

5.2 Dashboard & Courses (Protected)

Method	Endpoint	Description
GET	/api/dashboard/summary	Stats, streak, weekly chart, skill progress, achievements
GET	/api/dashboard/enrollments	Continue learning, all enrolled courses, recommended
GET	/api/courses	Full catalog with per-user enrolled flag
POST	/api/courses/{courseId}/enroll	Creates a real enrollment
GET	/api/courses/{courseId}/lessons	Lesson list with completion status and content
POST	/api/lessons/{lessonId}/complete	Marks a video/reading lesson complete
POST	/api/lessons/{lessonId}/submit-quiz	Grades quiz answers server-side

5.3 Account (Protected)

Method	Endpoint	Description
GET	/api/account/me	Full profile, enrollment stats, member-since date
PUT	/api/account/update	Edit personal details
POST	/api/account/change-password	Change password (blocked for Google-only accounts)

6. New & Modified Frontend Files

File	Purpose
ProtectedRoute.jsx	Route guard — redirects to login if not authenticated
Courses.jsx	Course catalog / browse & enroll page
CourseLearn.jsx	Lesson player: sidebar, video/reading/quiz views
ForgotPassword.jsx	Request a reset link
ResetPassword.jsx	Set a new password from the emailed link
Profile.jsx	View/edit account details, change password
Dashboard.jsx (rewritten)	Fetches real data from the backend instead of mock data

7. Notable Bugs Found & Fixed

- Google login silently failing after success — caused by React StrictMode double-invoking useEffect; fixed with a useRef guard.
- Navbar missing on Dashboard and Courses pages — fixed by rendering it on every protected page, not just the landing page.
- “Resume Trail” linking to an undefined course URL — fixed by including courseId in the dashboard API response.
- Seeding logic not idempotent — rewritten to find-or-create by title, safe to call repeatedly without losing real progress data.
- Backend restart requirement — several “fix isn't working” reports were resolved simply by restarting the Spring Boot process after Java changes.

8. Current Project Status

Area	Status
Google OAuth2 + JWT login, email/password	Complete & verified
Real dashboard (stats, streak, chart, achievements)	Complete & verified
Course catalog + real enrollment	Complete & verified

Area	Status
Real lessons — video, reading, quiz	Complete & verified
Server-graded quizzes	Complete & verified
Forgot / reset password via real email	Complete & verified
Signup personal details	Complete & verified
Account / profile settings page	Complete & verified
Merge into main via Pull Request	Not yet done
Cloud database migration	Deferred

9. Recommended Next Steps

- Email verification on signup — currently any email address can be used, even unverified ones.
- Search & filter on the course catalog — currently a flat, unsorted grid.
- Course preview page before enrolling — users currently enroll without seeing a syllabus.
- Rate limiting on login / forgot-password — no brute-force protection currently.
- Backend input validation — e.g. nothing currently blocks an empty password via a raw API call.
- Rotate exposed secrets — OAuth client secret, JWT secret, and mail credentials should all be rotated before production.
- Merge the feature branch into main via a reviewed Pull Request, since this is a team repository.